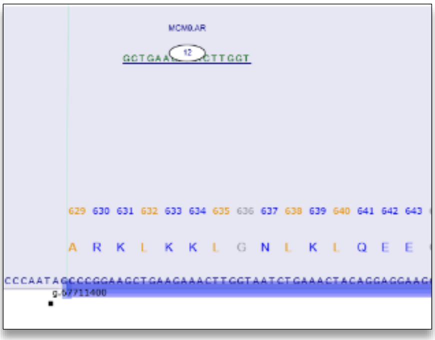
Mechanistically, these findings are consistent with the known properties of locked nucleic acids. The increased binding affinity conferred by LNA chemistry enhances target hybridization but also increases the likelihood of hybridization-dependent off-target effects and nonspecific interactions with cellular macromolecules. Both preclinical and clinical evidence suggest that such interactions may drive hepatic accumulation and toxicity, a phenomenon observed across multiple LNA-based oligonucleotides (Swayze et al., 2007; Bianchini et al., 2013). In the case of EZN-4176, its high-affinity binding profile likely contributed not only to hepatotoxicity but also to reduced specificity, limiting its ability to achieve selective and effective AR suppression in patients.

Importantly, although EZN-4176 was designed to target exon 4 of AR mRNA and suppress both full-length and splice variant receptors, clinical biopsy data failed to demonstrate meaningful AR downregulation following treatment (Bianchini et al., 2013). This disconnect between target engagement in preclinical models and clinical outcomes underscores a central challenge in antisense drug development: increasing binding affinity can improve in vitro potency but may also exacerbate off-target effects, narrow the therapeutic window, and ultimately hinder clinical translation. However, it should be noted that this compound also hits MCM9, a DNA repair gene linked to cancer. This seems to have been a bioinformatics oversight likely because the genome was not fully annotated at the time.

These limitations parallel observations from other antisense programs targeting the androgen receptor. For example, EZN-4176 can be contrasted with later-generation compounds such as ISIS-AR Rx (AZD5312), which incorporates some constrained ethyl (cEt) modifications designed to retain potency while improving tolerability and specificity. While cEt chemistry reduces some of the toxicity risks associated with LNA, clinical outcomes have shown that improved safety alone does not guarantee efficacy, particularly if sufficient target engagement or downstream biological effect is not achieved.

Taken together, these findings highlight a fundamental tension in antisense therapeutics: while increasing binding affinity (as seen with LNA chemistry) can enhance potency, it often comes at the cost of increased off-target interactions and toxicity. Conversely, newer chemistries such as cEt may improve safety margins but do not inherently ensure clinical success. Achieving an optimal balance between affinity, specificity, and tolerability remains a key determinant of successful antisense drug development.

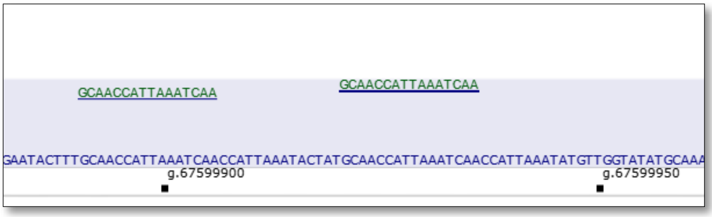
	EZN-4176	ISIS-AR Rx (AZD5312)
Developer	Enzon	Ionis / AstraZeneca
Chemistry	LNA gapmer	cEt gapmer
Target	Androgen receptor (AR)	Androgen receptor (AR)
Backbone	Phosphorothioate	Phosphorothioate
Sugar mods	LNA + DNA	cEt + DNA
Clinical outcome	ALT elevations (toxicity signal)	Developed to improve potency/safety



The Enzon–Santaris LNA gapmer program progressed into the clinic by targeting oncogenic drivers like AR and PI3K/Akt, but the same compound also unintentionally suppressed MCM9, a DNA repair gene linked to cancer predisposition. --a critical failure in bioinformatics off-target analysis that made it all the way into the clinic!

Feature	ISIS-AR Rx / AZD5312 (AKA ISIS 560131)
Base	TTGATTTAATGGTTGC
Backbone	Full phosphorothioate
Sugar	k k k d d d d d d d d d d k k E

Legend	k = (S)-cEt; d = DNA; E = 2′-MOE; C = 5-methylcytosine

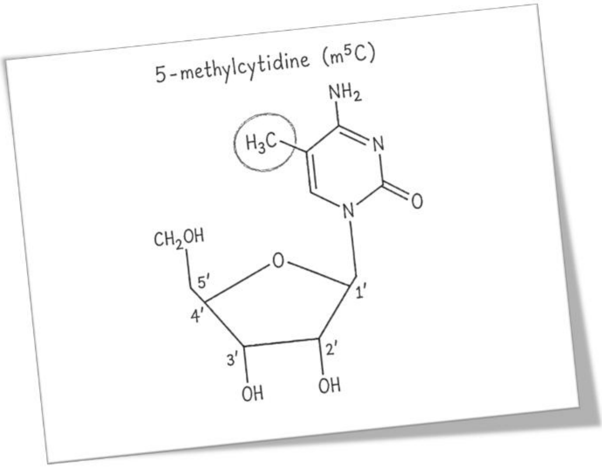


ISIS AR RX HITS ANDROGEN RECEPTOR TWICE.

Recent data (2024–early 2026) indicate that AZD5312 shows promising efficacy in prostate cancer with observed PSA declines in some heavily pre-treated patients and enhanced activity in combination with AKT inhibitors (e.g., Capivasertib). Its safety profile is consistent with antisense oligonucleotides, with common adverse events including fatigue, injection site reactions, and flu-like symptoms, while more serious but manageable risks include thrombocytopenia, hepatotoxicity, and renal effects requiring routine monitoring (AstraZeneca, 2024; ClinicalTrials.gov, 2025; de Bono et al., 2023).

Locked nucleic acid (LNA) chemistry is one of the most potent antisense modification strategies for RNA targeting due to its conformationally “locked” sugar, which markedly increases binding affinity and duplex stability relative to earlier chemistries such as 2′-O-methoxyethyl (MOE) and 2′-O-methyl modifications (Qassem et al., 2024; Crooke et al., 2018). This enhanced affinity enables shorter oligonucleotides, improved potency, and flexible designs (e.g., LNA/DNA gapmers) that retain RNase H activity while achieving stronger target engagement than MOE-based ASOs (Qassem et al., 2024). However, these advantages come with important trade-offs: compared with MOE chemistries, which generally exhibit lower toxicity and reduced immunogenicity, LNA-modified ASOs show a higher incidence of hybridization-dependent off-target effects due to their increased binding strength and shorter sequence length, as well as a greater risk of hepatotoxicity observed in preclinical models (Qassem et al., 2024; Crooke et al., 2018). Thus, while MOE modifications offer a more balanced safety profile with moderate potency, LNA chemistry shifts the balance toward maximal efficacy at the expense of a narrower therapeutic window, making careful sequence design and dose optimization critical for clinical application.

Nucleobase modifications



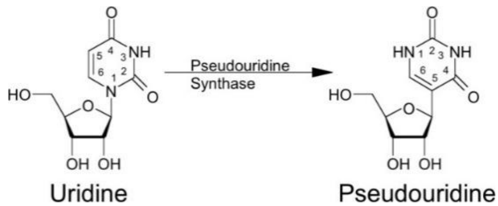
Chemical modification of RNA is essential for modern RNA therapeutics, enabling stability, reduced immunogenicity, enhanced target engagement, and greater pharmacological efficacy. Many of these modifications are inspired by naturally occurring post-transcriptional RNA modifications that regulate RNA structure and function in cells. By incorporating modified nucleosides or alternative backbone chemistries into therapeutic oligonucleotides, researchers can overcome many of the limitations associated with unmodified RNA, including rapid degradation, poor cellular persistence, and activation of innate immune responses. Among the most widely utilized modifications are 5-methylcytidine (m5C), pseudouridine (Ψ), inosine, and glycol nucleic acid (GNA), each contributing distinct structural and functional advantages.

5-Methylcytidine (m5C)

5-Methylcytosine (m5C; 5-methylcytidine, 5-Me-C) is a naturally occurring methylated RNA base found in tRNA and numerous other cellular RNAs, where it functions as a post-transcriptional modification that contributes to RNA stability, processing, and translation (Motorin and Helm, 2011; Squires et al., 2012). Its best-characterized role is in tRNA, where m5C protects transcripts from endonucleolytic cleavage and helps preserve structural integrity under cellular stress (Blanco et al., 2014).

These endogenous stabilizing properties have been leveraged in therapeutic RNA design. Incorporation of 5-methylcytidine into synthetic RNA oligonucleotides enhances base pairing and duplex stability through increased hydrophobicity at the C5 position, typically increasing melting temperature by approximately 1–1.5 °C per modification. Improved duplex stability strengthens target hybridization, facilitates disruption of target secondary structures, and can enable the use of shorter antisense oligonucleotides while maintaining binding affinity. In addition, m5C reduces susceptibility to nuclease degradation and attenuates activation of innate immune receptors that preferentially recognize unmodified RNA, thereby improving both stability and tolerability of RNA therapeutics (Karikó et al., 2005; Anderson et al., 2010).

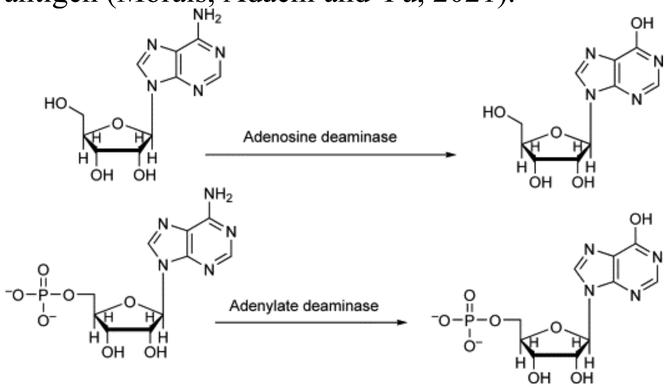
Pseudouridine (Ψ)



Pseudouridine (Ψ), the C5-glycoside isomer of uridine, is the most abundant naturally occurring RNA modification and is widely distributed across tRNA, rRNA, mRNA, and non-coding RNAs (Lin et al., 2021). Conversion of uridine to pseudouridine introduces a carbon–carbon glycosidic bond and an additional hydrogen-bond donor, enhancing base stacking interactions, strengthening RNA–RNA and RNA–protein interactions, and increasing structural rigidity. As a result, Ψ contributes to the stability of tRNA architecture, supports ribosomal function through rRNA modification, and regulates mRNA stability and translational efficiency (Zhao and He, 2015; Lin et al., 2021).

These properties have made pseudouridine one of the most important modifications in RNA therapeutics. Incorporation of Ψ into synthetic RNAs enhances duplex stability, improves resistance to nuclease degradation, and increases translational output, often resulting in substantially higher protein expression compared with unmodified RNA (Lin et al., 2021). Equally important, pseudouridine reduces activation of innate immune sensors and downstream cytokine responses, thereby enabling sustained translation and improved tolerability of exogenous RNA (Karikó et al., 2008; Lin et al., 2021).

The clinical significance of pseudouridine is exemplified by mRNA COVID-19 vaccines. These vaccines consist of a nucleoside-modified mRNA encoding the SARS-CoV-2 spike protein, encapsulated within a lipid nanoparticle (LNP) delivery system. The mRNA contains regulatory elements including 5′ and 3′ untranslated regions, a coding sequence encoding a stabilized pre-fusion spike protein, and multiple stop codons to ensure accurate translation (Morais, Adachi and Yu, 2021). A defining feature of these vaccines is the substitution of uridine with N1-methyl-pseudouridine, which enhances mRNA stability, suppresses innate immune activation, and improves protein translation efficiency (Morais, Adachi and Yu, 2021). The surrounding LNP protects the mRNA from degradation, facilitates cellular uptake, and promotes delivery to antigen-presenting cells such as dendritic cells, ultimately enabling robust immune responses against the encoded antigen (Morais, Adachi and Yu, 2021).



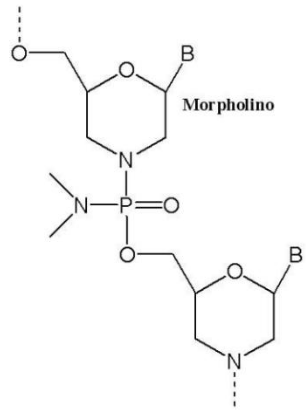
Inosine

Inosine is a functionally unique RNA modification generated through adenosine-to-inosine (A-to-I) editing by adenosine deaminase acting on RNA (ADAR) enzymes. Because inosine is interpreted by cellular machinery as guanosine, A-to-I editing enables dynamic modification of RNA information without altering the underlying genome. This process can recode proteins, alter splice-site selection, and modify RNA secondary structure, thereby influencing transcript stability, localization, and protein expression (Eisenberg and Levanon, 2018; Srinivasan et al., 2021).

In naturally occurring tRNAs, inosine located at the wobble position expands codon recognition, allowing a single tRNA to decode multiple synonymous codons and thereby enhancing translational efficiency and proteome diversity (Srinivasan et al., 2021). In mRNAs, site-specific editing can produce amino acid substitutions or regulate RNA processing, with important roles in neurological function and development.

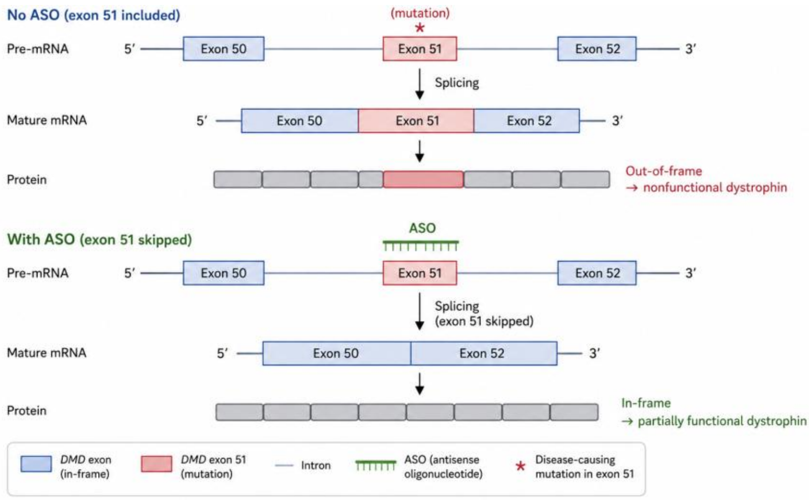
These biological functions have provided the foundation for RNA editing therapeutics. Programmable A-to-I editing exploits endogenous ADAR enzymes to correct disease-causing mutations at the RNA level, offering a reversible and potentially safer alternative to permanent genome editing. In addition, inosine-containing RNAs exhibit reduced activation of innate immune sensors such as MDA5, improving the safety profile of therapeutic RNAs. Its flexible base-pairing properties also provide opportunities to modulate RNA structure and translation, making inosine relevant to the design of antisense oligonucleotides, siRNAs, and mRNA therapeutics. Despite its promise, challenges including off-target editing, variable editing efficiency, and delivery limitations continue to constrain broader clinical application.

Morpholinos



The development of morpholino oligonucleotides arose from the need to overcome fundamental limitations of early antisense chemistries, particularly the poor stability of natural oligonucleotides in biological systems due to nuclease degradation and the nonspecific interactions caused by their negatively charged phosphodiester backbone (Stein and Cheng, 1993; Summerton, 1999). To address these issues, James Summerton in 1985 working at Antivirals which later became Santarus Inc replaced the ribose sugar with a chemically robust morpholine ring and substituted the charged phosphodiester linkage with a neutral phosphorodiamidate backbone. (Summerton and Weller, 1997; Li, 2017, p. 40). This elegant design preserved the Watson–Crick base pairing and allowed for effective sequence-specific binding to target RNA and subsequent steric blocking for transcript manipulation (Moulton and Moulton, 2017, pp. v–vi, 11).

In the early 1990s, Zbigniew Dominski and Ryszard Kole at the University of North Carolina demonstrated that targeting specific splice elements in β-globin pre-mRNA could redirect the splicing machinery and restore correct mRNA production in vitro. This seminal work laid the foundation for the modern splice-switching therapeutic strategies we have today, showing that RNA splicing is not only mechanistically tractable but also chemically programmable. The favorable safety of morpholinos subsequently enabled the first clinical applications of splicing therapeutics.



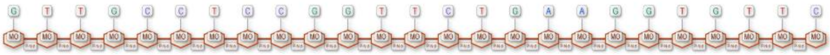
Researchers at Kobe University first demonstrated that ASOs could redirect splicing of the dystrophin transcript to induce exon skipping, laying the foundation for modern RNA therapeutics for Duchenne muscular dystrophy (DMD).

Takeshima Y, et al. (1995). "Modulation of in vitro splicing of the upstream intron by modifying an intra-exon sequence in the dystrophin gene."
Journal of Clinical Investigation 95(2): 515–520.
DOI: 10.1172/JCI117693

AVI-4658, an exon 51–targeting PMO, showed in a proof-of-concept clinical study showed that direct intramuscular administration could induce sequence-specific exon skipping and restore dystrophin expression locally in patients, without significant safety concerns (Kinali et al., 2009). This study provided the first clinical validation of PMOs as a therapeutic platform, establishing proof-of-mechanism in humans and laying the foundation for subsequent systemic PMO-based therapies.

Eteplirsen

Eteplirsen, a phosphorodiamidate morpholino oligomer, was evaluated in a small randomized, placebo-controlled trial in boys with Duchenne muscular dystrophy amenable to exon 51 skipping, demonstrating that sustained treatment restores production of truncated but functional dystrophin (Mendell et al., 2013). While no meaningful dystrophin increase was observed at 12 weeks, significant increases in dystrophin-positive muscle fibers were seen by 24 weeks and reached 40-50% by 48 weeks, with corresponding restoration of associated proteins such as nNOS and sarcoglycans (Mendell et al., 2013). Functionally, Eteplirsen-treated patients showed stabilization of walking ability compared with the expected decline in placebo patients, with an approximate 67 m benefit in the 6-minute walk test among evaluable patients (Mendell et al., 2013). The study suggests that treatment duration, rather than dose, is the key driver of efficacy, and that Eteplirsen is generally well tolerated, though conclusions are limited by the small sample size and variability in disease progression (Mendell et al., 2013).



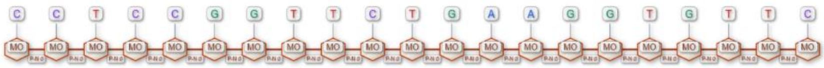
Golodirsen (Vyondys 53)

Golodirsen (Vyondys 53) is a phosphorodiamidate morpholino oligomer (PMO) antisense oligonucleotide indicated for Duchenne muscular dystrophy (DMD) patients with mutations amenable to exon 53 skipping. It binds to exon 53 of dystrophin pre-mRNA, promoting exon skipping and enabling production of a truncated but partially functional dystrophin protein (FDA, 2019). It received accelerated approval in the U.S. based on increased dystrophin production rather than confirmed clinical outcomes (FDA, 2019). The recommended dosing is 30 mg/kg IV weekly (FDA, 2019).

Exon skipping	Increased in all patients; 28.9-fold increase at 48 weeks (FDA, 2019; Servais et al., 2022)
Dystrophin	0.10% → 1.02% of normal (mean +0.92%)
Dose	30 mg/kg IV weekly
PK linearity	Dose-proportional exposure; minimal accumulation
Cmax variability	38–72% CV
AUC variability	34–44% CV
Vd (steady-state)	668 mL/kg
Protein binding	33–39%
Half-life	3.4 hours
Clearance	346 mL/hr/kg

Metabolism	Minimal; no metabolites detected
Excretion	Primarily unchanged in urine
Renal impairment	AUC ↑ 1.2× (Stage 2 CKD), 1.9× (Stage 3 CKD) (FDA, 2019)
DDI potential	Low (FDA, 2019)

Viltolarsen (Viltepso)



Viltolarsen (Viltepso) is a phosphorodiamidate morpholino oligomer that induces exon 53 skipping in dystrophin pre-mRNA, restoring production of a truncated but functional dystrophin protein (FDA, 2020). Treatment results in measurable increases in dystrophin expression (~5–6% of normal) and consistent exon skipping at the mRNA level, confirming target engagement (FDA, 2020; Clemens et al., 2020). Pharmacokinetically, viltolarsen demonstrates dose-proportional exposure, minimal accumulation, and rapid renal elimination of unchanged drug, with low metabolic and drug–drug interaction potential (FDA, 2020). Clinically, it is generally well tolerated, with mostly mild adverse events (e.g., respiratory infections, injection-site reactions), no serious drug-related adverse events in trials, and low immunogenicity; however, renal toxicity remains a theoretical class risk, requiring ongoing monitoring (FDA, 2020; Clemens et al., 2020).

Viltolarsen is generally well tolerated, with most adverse events being mild and including upper respiratory tract infections, cough, pyrexia, and injection-site reactions (FDA, 2020). Clinical trial data show no serious drug-related adverse events or treatment discontinuations, supporting an overall favorable safety profile (Clemens et al., 2020). However, as with other antisense oligonucleotides, there is a potential risk of renal toxicity, necessitating routine monitoring during treatment (FDA, 2020). Immunogenicity appears minimal, with no significant immune responses reported (FDA, 2020).

Administration	IV infusion (60 min)
Bioavailability	100%
Tmax	1 hour (end of infusion)
Exposure	Dose-proportional (1.25–80 mg/kg/week)
Accumulation	Minimal with weekly dosing
Volume of distribution	300 mL/kg
Plasma protein binding	39–40%
Metabolism	Metabolically stable; no metabolites detected
Elimination	Primarily renal (unchanged drug)
Half-life (t½)	2.5 hours
Clearance	217 mL/hr/kg
Drug interactions	Low potential (no major CYP/transporter effects)

Casimersen (AMONDYS 45)

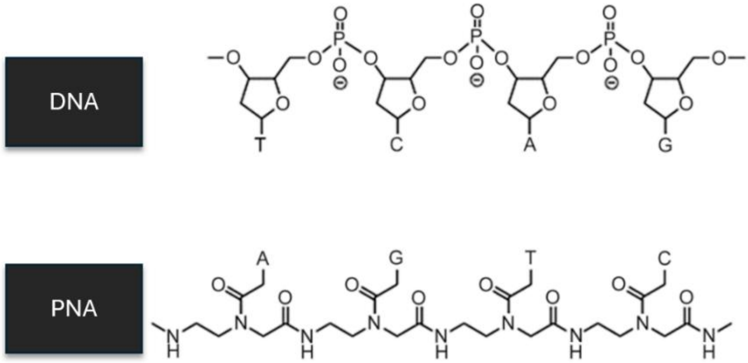
Casimersen (AMONDYS 45) is a phosphorodiamidate morpholino oligomer (PMO) antisense oligonucleotide approved for the treatment of Duchenne muscular dystrophy (DMD) in patients with mutations amenable to exon 45 skipping (FDA, 2021). DMD is an X-linked recessive disorder characterized by absence of functional dystrophin, leading to progressive degeneration and premature mortality (Assefa et al., 2024). Casimersen received accelerated FDA approval based on its ability to increase dystrophin production, a surrogate biomarker, with confirmatory trials required to establish clinical benefit (FDA, 2021).

Casimersen acts at the level of pre-mRNA splicing. It binds specifically to exon 45 of dystrophin pre-mRNA, preventing its inclusion during mRNA processing. This restores the translational reading frame and allows synthesis of a truncated but partially functional dystrophin protein (FDA, 2021; Assefa et al., 2024). In a placebo-controlled study dystrophin increased from 0.93% to 1.74% of normal after 48 weeks

Casimersen is generally well tolerated, with most adverse events being mild to moderate and consistent with common intercurrent illnesses. The most frequently reported adverse reactions include upper respiratory tract infection (65%), cough (33%), pyrexia (33%), headache (32%), arthralgia (21%), and oropharyngeal pain (21%) (FDA, 2021). Despite this favorable tolerability profile, an important safety concern is the potential for renal toxicity, which has been observed in animal studies and is considered a class effect of antisense oligonucleotides. Consequently, careful renal monitoring is recommended during treatment, including assessment of serum cystatin C, urine protein, and urine protein-to-creatinine ratio to detect early signs of nephrotoxicity (FDA, 2021) (Assefa et al., 2024).

Key PD outcome	+0.81% dy`strophin vs +0.22% placebo
Dose	30 mg/kg IV weekly
Cmax	End of infusion
Half-life	3.5 h
Clearance	180 mL/hr/kg
Vd	367 mL/kg
Protein binding	8.4–31.6%
Metabolism	Minimal (non-hepatic)
Excretion	>90% renal (unchanged)
Accumulation	None
Key safety risk	Renal toxicity
Common AEs	URTI, cough, pyrexia, headache

Peptide Nucleic Acids (PNAs)



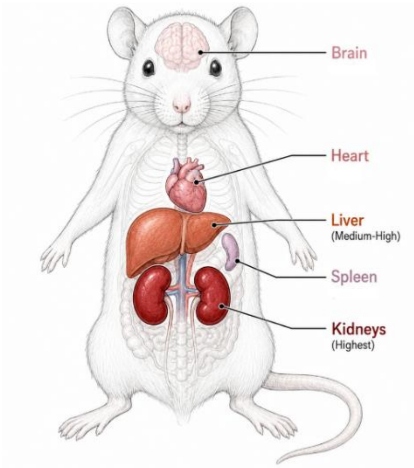
Peptide nucleic acids (PNAs) are synthetic DNA/RNA analogues in which the negatively charged sugar-phosphate backbone of natural nucleic acids is replaced by a neutral pseudopeptide backbone composed of N-(2-aminoethyl)-glycine units. The nucleobases adenine, guanine, cytosine and thymine are attached to this backbone through methylene carbonyl linkers, allowing PNAs to hybridize with complementary DNA or RNA through Watson-Crick base pairing. This distinctive backbone chemistry gives PNAs unusually strong binding affinity, high thermal stability and resistance to enzymatic degradation, making them attractive tools for gene targeting, antisense inhibition and sequence-specific nucleic acid recognition (Nielsen et al., 1991; Egholm et al., 1993; Demidov et al., 1994).

Since their discovery by Peter Nielsen and colleagues at the University of Copenhagen and Risø National Laboratory in Denmark in 1991, PNAs have evolved into promising therapeutic candidates. Early studies established that PNAs bind nucleic acids in a sequence-selective manner and often form more stable duplexes than comparable DNA/RNA systems (Nielsen et al., 1991; Egholm et al., 1993). This enhanced stability is largely attributed to the neutral PNA backbone, which avoids the electrostatic repulsion that occurs between negatively charged DNA or RNA strands. As a result, PNA/DNA hybrids typically exhibit substantially higher melting temperatures than equivalent DNA/DNA duplexes, with reported increases of approximately 7–20 °C under comparable experimental conditions (Schwarz, Robinson and Butler, 1999). However, PNA/DNA duplex stability remains dependent on sequence, length and base composition, meaning that Tm values must be interpreted in the context of the specific oligonucleotide and assay conditions used (Giesen et al., 1998).

These properties make PNAs particularly useful as antisense agents. By binding to target mRNA, PNAs can block gene expression by preventing translation or interfering with other nucleic-acid-dependent processes. In antibacterial applications, early studies showed that PNAs could inhibit bacterial gene expression, especially when conjugated to cell-penetrating peptides that improve transport across bacterial membranes (Good et al., 2001). More recent work has expanded this strategy to target essential bacterial genes, resistance genes and virulence genes, positioning PNAs as programmable, narrow-spectrum antimicrobial agents with potential value against antibiotic-resistant pathogens (El-Fateh, Chatterjee and Zhao, 2024).

Compared with other nucleic-acid-based therapeutic platforms, PNAs are distinguished by their neutral pseudopeptide backbone, high target affinity and resistance to nuclease and protease degradation. DNA antisense oligonucleotides, particularly phosphorothioate-modified ASOs, retain an anionic character while gaining improved nuclease stability and increased interactions with plasma and cellular proteins. Morpholinos, or phosphorodiamidate morpholino oligomers, also have neutral backbones, but they use morpholine rings and phosphorodiamidate linkages rather than the PNA pseudopeptide scaffold. Like PNAs, morpholinos are metabolically stable and primarily act through steric blocking; unlike many DNA gapmer ASOs, they do not recruit RNase H to degrade target RNA (Heo et al., 2024). These differences in backbone chemistry help explain the distinct binding, stability, pharmacokinetic and pharmacodynamic profiles of PNA, DNA antisense oligonucleotides and morpholinos.

The plasma pharmacokinetic profiles of these chemistries are also markedly different. Unmodified PNA appears to clear rapidly from plasma; in a rat intravenous study, a PNA targeting the rat neurotensin receptor had a distribution half-life of approximately 3 minutes and an elimination half-life of approximately 17 minutes (McMahon et al., 2002). Phosphorothioate DNA ASOs generally show more prolonged systemic exposure and extensive tissue distribution, driven largely by plasma protein binding and tissue uptake (Geary et al., 2015). Morpholinos such as eteplirsen also have relatively short plasma half-lives compared with many phosphorothioate ASOs; eteplirsen has a reported elimination half-life of approximately 3–4 hours, with renal clearance accounting for a substantial fraction of the administered dose within 24 hours (FDA, 2016).



Protein binding is one of the clearest differentiators between phosphorothioate DNA ASOs and neutral-backbone chemistries. Phosphorothioate ASOs interact extensively with proteins, and these interactions affect distribution, tissue delivery, cellular uptake, intracellular trafficking, potency and toxicity. PNA and morpholinos, because they are neutral molecules, generally show lower nonspecific electrostatic protein interactions than phosphorothioate ASOs. This lower protein binding can reduce plasma retention and contribute to more rapid renal clearance, particularly for unconjugated neutral oligonucleotide analogues (McMahon et al., 2002; Heo et al., 2024).

Although PNA, DNA ASOs and morpholinos all distribute beyond plasma, their dominant tissue patterns differ. In the rat PNA study, low but detectable PNA levels were observed across examined organs, with the highest concentrations in kidney, followed by liver, heart, brain and spleen (McMahon et al., 2002). Phosphorothioate DNA ASOs generally show broad systemic biodistribution, with highest tissue concentrations commonly observed in liver and kidney, followed by other tissues such as bone marrow, adipose tissue and lymph nodes (Geary et al., 2015). Morpholinos also show class-consistent distribution and elimination properties, with approved PMOs such as eteplirsen, golodirsen and casimersen sharing broadly similar ADME/DMPK characteristics (Heo et al., 2024).

Despite their strong hybridization properties, PNAs have poor intrinsic cellular uptake, so their pharmacodynamic activity is often limited by delivery rather than target binding once inside the cell. Phosphorothioate ASOs are taken up mainly through endocytic pathways, although only a fraction of internalized oligonucleotide typically reaches productive intracellular compartments. Morpholinos face a similar delivery challenge,